Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Physical vector spaces for quantum mechanics

Files.

- Files: `briefs/QuantumVectorSpaces/QuantumVectorSpaces.lean`, and `briefs/QuantumVectorSpaces/QuantumVectorSpaces/Countable.lean` (a countable generating set of observables, Sec. 5), and `briefs/QuantumVectorSpaces/QuantumVectorSpaces/PureStates.lean` (second countability from the pure states, Sec. 5), and `briefs/QuantumVectorSpaces/QuantumVectorSpaces/Frames.lean` (frame changes, Theorem 5, [B]WDQ), and `briefs/QuantumVectorSpaces/QuantumVectorSpaces/Isomorphisms.lean` (isomorphisms of quantum vector spaces, Example 3 and Theorem 5)
- Library: `QuantumVectorSpaces`
- Imports: `SchwartzNorms` (the L^2 seminorm family on $\mathcal{S}(\mathbb{R})$) and new `ForMathlib` files: `ForMathlib/Analysis/InnerProductSpace/Defs.lean` (the seminorm $\sqrt{\mathrm{re}\langle x, x\rangle}$ of a `PreInnerProductSpace.Core`, and continuity of the inner product against it), `ForMathlib/Analysis/CStarAlgebra/Matrix.lean` (`Matrix.toEuclideanLin` is

multiplicative),
`ForMathlib/Analysis/Distribution/SchwartzSpace/Complete.lean`
(`SchwartzMap.instCompleteSpace`: $\mathcal{S}(E, F)$ is complete for complete F),
`ForMathlib/Analysis/Distribution/SchwartzSpace/SupNorm.lean`
(`SchwartzMap.exists_norm_le_toLp_two`: a Sobolev bound for the values of a Schwartz function, from TauCeti's `SchwartzMap.exists_integral_norm_fourierInv_le`),
`ForMathlib/Analysis/InnerProductSpace/MultilinearBasis.lean`
(`ContinuousMultilinearMap.norm_le_sum_orthonormalBasis`: a multilinear map is bounded by its values on an orthonormal basis),
`ForMathlib/Analysis/LocallyConvex/FirstCountable.lean`
(`WithSeminorms.exists_countable`: a family of seminorms inducing a first-countable topology has a countable subfamily inducing it),
`ForMathlib/Topology/Algebra/Projectivization/SecondCountable.lean`
(`Projectivization.secondCountableTopology_of`: with a separating dual, V is second countable if its projective space is)

Setting. V is a complex vector space with an inner product given as a `PreInnerProductSpace.Core`. Unlike Mathlib's `InnerProductSpace`, this puts no norm or topology on V. The defining observables are a family X_i ($i \in \iota$) in `Module.End`, and `algebra` is the unital algebra they generate (`Algebra.adjoin`). The seminorms $p_A(v) = \sqrt{\langle Av, Av\rangle}$ form `seminormFamily`, and `topology` is the topology they generate. A topology is `IsAdmissible` if it makes V a topological vector space and makes the X_i and the inner product (jointly) continuous. For Example 3, $Xf = xf$ and $Pf = -i\hbar f'$ act on Mathlib's $\mathcal{S}(\mathbb{R}, \mathbb{C})$ (`posOp`, `momOp`), with the L^2 inner product `l2Core`. For Example 4, $S_i = (\hbar/2)\sigma_i$ acts on $\mathbb{C}^2$ as `EuclideanSpace` (`spin`). In n dimensions, $X^i f = x_i f$ and $P_i f = -i\hbar\,\partial_{e_i} f$ act on $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$, with $\mathbb{R}^n$ as `EuclideanSpaceâDİ(Finn)` (`posOpN`, `momOpN`, indexed by `ObsNn`), and the L^2 inner product `l2CoreN`.

Definition 1 is `IsQuantumVectorSpacecXÄğC`: the X_i are symmetric, they satisfy $[X_i, X_j] = i\hbar \sum_k C_{ijk} X_k$ with finitely many nonzero C_{ijk} for each i, j, and V with `topology` is complete and second-countable. Completeness refers to `uniformSpace`, the uniform structure that every topological vector space carries.

Results.

Brief	Lean
Corollary 2, coarsest topology	`isGreatest_topology` (from `isAdmissible_topology` and `IsAdmissible.continuous_algebra`)
Corollary 2, locally convex	`locallyConvexSpace`
finite dimension (Example 4)	`topology_eq_moduleTopology`, `topology_qubit`
Example 3, $[X, P] = i\hbar$	`posOp_mul_momOp_sub`
Example 3, X, P, I symmetric	`isSymmetric_obs`
Example 3, Remark: the topology is that of $\mathcal{S}$	`topology_schwartz`, `withSeminorms_schwartz`, `isAdmissible_schwartz` (for $\hbar \neq 0$)
Example 4, $[S_i, S_j] = i\hbar\,\epsilon_{ijk} S_k$	`spin_comm`
Example 4, S_i symmetric	`isSymmetric_spin`
Example 4 is a quantum vector space	`isQuantumVectorSpace_qubit`
Example 3, second countability	`secondCountableTopology_schwartz`

Brief	Lean
Example 3, completeness	`SchwartzMap.instCompleteSpace`
Example 3 is a quantum vector space	`isQuantumVectorSpace_schwartz` (for $\hbar \neq 0$)
Example 3 in n dimensions, commutation relations	`posOpN_comm`, `momOpN_comm`, `posOpN_comm_momOpN`
Example 3 in n dimensions, symmetric observables	`isSymmetric_obsN`
Example 3 in n dimensions, Remark: the topology is that of $\mathcal{S}$	`topology_schwartzN` (for $\hbar \neq 0$)
Example 3 in n dimensions, second countability	`secondCountableTopology_schwartzN`
Example 3 in n dimensions is a quantum vector space	`isQuantumVectorSpace_schwartzN` (for $\hbar \neq 0$)
Theorem 5, [B]COBS	`IsAdmissible.continuous_expectation`
Sec. 5, a countable subset of observables generates the topology	`exists_countable_topology_eq`, `exists_countable_topology_eq_iff` (under first countability)
Sec. 5, the same for a quantum vector space	`IsQuantumVectorSpace.exists_countable`
Sec. 5, separability is not enough	`Diagonal.separableSpace`, `Diagonal.not_firstCountableTopology`, `Diagonal.not_exists_countable`
Sec. 5, second countability from the pure states	`secondCountableTopology_iff_projectivization` (for a definite inner product)
Sec. 5, the pure states of a quantum vector space are second countable	`IsQuantumVectorSpace.secondCountableTopology_projectivization`
Theorem 5, [B]WDQ: two frames with each other's defining observables as observables have the same observables	`topology_eq_iff_continuous`, `isObservable_iff_of_frames`
Theorem 5, [B]WDQ: sufficient conditions	`topology_eq_of_mem_algebra`, `topology_eq_of_finiteDimensional`
Theorem 5, [B]WDQ: the premise is not automatic	`Diagonal.not_continuous_mulOp`
Isomorphisms of quantum vector spaces	`UnitaryEquiv`, `LieIso` (with `refl`, `symm`, `trans`), `LieIso.toUnitaryEquiv`
Isomorphisms preserve observables and pure states	`UnitaryEquiv.isObservable_conj_iff`, `UnitaryEquiv.pureStates`
Isomorphisms and frame changes	`induced_topology_eq_pullback`, `induced_topology_eq_iff`

Differences from the brief.

- "Coarsest" is `IsGreatest` in Mathlib's order on topologies, where coarser is larger.
- The inner product is only assumed positive semidefinite in Corollary 2; definiteness is used only in finite dimension, to make the topology Hausdorff.
- Example 3 is formalized twice. For $\mathcal{S}(\mathbb{R})$, the proof that the topologies agree has two

halves. (a) Each seminorm $\|x^a\partial^b f\|_2$ equals $|\hbar|^{-b}p_{X^aP^b}(f)$. (b) Every element of the algebra is a combination of the normally ordered monomials X^aP^b (`algebra_le_monoSpan`, from $PX^{n+1} = X^{n+1}P - (n+1)i\hbar X^n$), so every p_A is bounded by finitely many $\|x^a\partial^b f\|_2$. With `withSeminorms_l1_l2` from the previous brief, this gives the Schwartz topology.

- For $\mathcal{S}(\mathbb{R}^n)$, one half is that the Schwartz topology is admissible (`isAdmissible_schwartzN`), hence finer than the topology of Corollary 2. For the other half, each Schwartz seminorm $\sup_x \|x\|^k\|D^m f(x)\|$ is bounded by finitely many p_A (`exists_seminorm_le`). The derivative $D^m f(x)$ is bounded by its values on the unit vectors, $\|x\|^k \le (1+\|x\|^2)^k$ is the polynomial $(1+\sum_i (X^i)^2)^k$, and the Sobolev bound $|g(x)| \le C(\|g\|_2 + \sum_i \|\partial_i^{n+1} g\|_2)$ turns the supremum into L^2 norms of elements of the algebra applied to f. No normal ordering is needed, since the operators $\partial_i^{n+1}(1+\sum_j (X^j)^2)^k\partial^r$ are themselves in the algebra.

- In finite dimension the topology of Corollary 2 is the unique Hausdorff vector-space topology, whatever the defining observables (`topology_eq_moduleTopology`). This is how Example 4 is checked.

- Second countability in n dimensions comes from a general fact (`secondCountableTopology_topology`). With countably many defining observables, the p_A for the words $A = X_{i_1}\cdots X_{i_k}$ already generate the topology, so $v \mapsto (Av)_A$ embeds V into a countable product of copies of L^2.

- Second countability of $\mathcal{S}(\mathbb{R})$ is proved by embedding it into a countable product of copies of $L^2(\mathbb{R})$ through $f \mapsto (x^a\partial^b f)_{a,b}$ (`l2Embedding`). The topology of $\mathcal{S}(\mathbb{R})$ is induced by this map, because it is given by the L^2 seminorms (`withSeminorms_l1_l2`).

- Completeness of $\mathcal{S}(E,F)$, for any complete F, is proved in `ForMathlib`, since neither Mathlib (at this pin or on master) nor TauCeti nor physlib has it. The proof is the classical one (Becnel and Sengupta, *Mathematics* 3 (2015), §3.5): along a Cauchy sequence the weighted derivatives $\|x\|^k D^n u_m$ converge uniformly; the limits are the derivatives of the limit function, by the theorem on uniform limits of derivatives (`hasFDerivAt_of_tendstoUniformly`); and the bounds pass to the limit. For the qubit, completeness and second countability come from finite dimension.

- The claim of Sec. 5 that a countable subset of the observables generates the topology is proved under first countability, which second countability implies; it is in fact equivalent to first countability (`exists_countable_topology_eq_iff`). The proof (`WithSeminorms.exists_countable`) takes a countable basis of balls at 0; each ball involves finitely many words $X_{i_1}\cdots X_{i_k}$, and their letters form the countable subset. For a quantum vector space the subset is enlarged to be closed under the structure constants, so that it satisfies Definition 1 itself (`IsQuantumVectorSpace.exists_countable`).

- Separability is not enough. On the finitely supported sequences, with the ℓ^2 inner product and every multiplication operator $(M_f v)_n = f_n v_n$ by a real sequence f as a defining observable, the observables are symmetric and commute (`Diagonal.isSymmetric_mulOp`, `Diagonal.mulOp_comm`), and the space is separable, as the union of countably many finite-dimensional subspaces (`Diagonal.separableSpace`). But the topology is not first countable: for countably many balls U_n at 0, choose $t_n > 0$ with $t_n e_n \in U_n$; then $\|M_f(t_n e_n)\| = 1$ for $f_n = 1/t_n$, so no U_n lies in the ball of p_{M_f} (`Diagonal.not_firstCountableTopology`). Hence no countable subset of the observables generates the topology (`Diagonal.not_exists_countable`). The topology is the finest locally convex one, so the space is also complete; this is not formalized.

- The pure states are the rays of V, that is, the projective space $\mathbb{P}(V)$. Its topology is the quotient topology from the nonzero vectors, which is Mathlib's `Projectivization.instTopologicalSpace`. With a definite inner product, V is second countable if and only if $\mathbb{P}(V)$ is (`secondCountableTopology_iff_projectivization`), so the claim of Sec. 5 holds with the topology of Definition 1 itself; nothing has to be extended. One direction holds because the quotient map is open. For the other, on the open set where $\langle v_0, v\rangle \neq 0$ the map $v \mapsto ([v], \langle v_0, v\rangle)$ is an embedding into the second-countable space $\mathbb{P}(V) \times \mathbb{C}$ (`Projectivization.isInducing_chart`); countably many such sets cover the nonzero vectors, because $\mathbb{P}(V)$ is Lindelöf, and a translate of one of them covers 0. The topology that the ensembles would induce on the pure states is not compared with the quotient topology, since the brief defines no ensemble space over V.

- For [B]WDQ, two frames are two families of defining observables X_i^A and X_j^B on the same space V with the same inner product, and an observable of a frame is a symmetric operator that is continuous for its topology (`IsObservable`). The brief's premise, that the defining observables of each frame are observables in the other, is a physical assumption and is kept as a hypothesis. It holds if and only if the two frames give the same topology (`topology_eq_iff_continuous`): the topology of B is then admissible for A, hence finer than that of A by Corollary 2, and vice versa. The two frames then have the same observables (`isObservable_iff_of_frames`), so the identity is a linear homeomorphism that preserves the inner product and all observables. The premise holds when the defining observables of each frame lie in the algebra of the other (`topology_eq_of_mem_algebra`), and always in finite dimension (`topology_eq_of_finiteDimensional`). It does not follow from sharing the space and the inner product: on the finitely supported sequences, the multiplication by n, a defining observable of the frame with all multiplication operators, is not continuous for the frame whose only defining observable is the identity (`Diagonal.not_continuous_mulOp`), since $\|e_n\| = 1$ while $\|ne_n\| = n$.

- The brief does not define isomorphisms of quantum vector spaces. Two notions are formalized, each closed under identity, inverse and composition. A *unitary equivalence* (`UnitaryEquiv`) is a linear bijection $U\colon V \to W$ that preserves the inner product and carries the topology of W to that of V. Conjugation $A \mapsto UAU^{-1}$ is then a bijection of the observables (`UnitaryEquiv.isObservable_conj_iff`), U induces a homeomorphism of the pure states (`UnitaryEquiv.pureStates`), and the Born rule is preserved with the inner product. Along any U that preserves the inner product, the topology of W pulls back to the topology on V with defining observables $U^{-1}Y_jU$ (`induced_topology_eq_pullback`). So U is a unitary equivalence if and only if the $U^{-1}Y_jU$ are continuous on V and the UX_iU^{-1} are continuous on W (`induced_topology_eq_iff`); for $U = \mathrm{id}$ this is the frame condition above. A *Lie isomorphism* (`LieIso`) is a linear bijection that preserves the inner product and whose conjugation maps the complex Lie algebra spanned by the X_i (`lieAlgebra`), with the commutator as bracket, onto that spanned by the Y_j. This is the brief's "preserve the Lie algebraic structure". Every Lie isomorphism is a unitary equivalence (`LieIso.toUnitaryEquiv`): the two Lie algebras generate conjugate associative algebras, hence the same topology.

- The "iff $n = m$" part of Example 3 is not formalized, and its truth depends on the notion of isomorphism. For Lie isomorphisms and $\hbar \neq 0$ it holds: X^i, P_i and I are linearly independent and span a Lie algebra of dimension $2n + 1$, and conjugation preserves the dimension. For unitary equivalences it fails. The Hermite functions h_α ($\alpha \in \mathbb{N}^n$) form an orthonormal basis of $L^2(\mathbb{R}^n)$, and f lies in $\mathcal{S}(\mathbb{R}^n)$ exactly when its coefficients c_α satisfy $\sum_\alpha |c_\alpha|^2(1+|\alpha|)^{2k} < \infty$

for all k, these sums giving the topology (Reed and Simon, *Methods of Modern Mathematical Physics I*, Thm. V.13, for $n = 1$; B. Simon, *J. Math. Phys.* 12 (1971) 140). Listing the α by total degree, the j-th one has $|\alpha| \sim j^{1/n}$, so both weight families define the space s of rapidly decreasing sequences. Hence $\mathcal{S}(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^m)$ are unitarily equivalent for all $n, m \geq 1$.

- Not formalized: [B]STDIST in Theorem 5 and the other remarks of section 5, which are informal.

Problems found in the brief. One claim is false, two are imprecise, and two steps are left implicit.

- Sec. 5 says that "separability/second countability would allow us to find a countable subset of observables from a bigger set that generates the same topology". This holds for second countability (first countability suffices), but not for separability: see the counterexample under "Differences".

- Sec. 5 says that "if the pure states are second countable, we can always extend the topology to the vector space to preserve second countability". It does not say which topology the pure states carry or what extending means. With the quotient topology, no choice of topology on V is involved: the topology of Definition 1 is second countable if and only if the pure states are.

- Example 3 says that $\mathcal{S}(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^m)$ "are isomorphic as quantum vector spaces if and only if $n = m$", but the brief does not define these isomorphisms. The argument of the Remark, that "no linear transformation can change the number of defining observables", is not valid: the number of defining observables is not an invariant. A unitary equivalence $\mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R})$ carries the $2n+1$ defining observables X^i, P_i, I to defining observables on $\mathcal{S}(\mathbb{R})$ that give the same topology. The claim holds for Lie isomorphisms and fails for unitary equivalences (see "Differences").

- The Remark after Example 3 says the L^2 seminorms and the sup seminorms dominate each other. But the seminorms of Corollary 2 are $\|Af\|$ for every A in the algebra, including non-normally-ordered products such as PX. Reducing them to $\|x^a \partial^b f\|$ needs the commutation relation (normal ordering), which the brief does not mention.

- Example 3 needs $\hbar \neq 0$: for $\hbar = 0$ we get $P = 0$, and the topology is just that of L^2.